

PX8-20H-01 DATASHEET

(Piezo Experimental, 8 nozzles, 20mm, High Voltage, Build 01)

Description: An experimental print head making use of 8 20mm PZT Piezoelectric Ceramic Buzzers to fire ink droplets onto a page

Manufacturer

Chowder INC
(Open Source Hardware)

ELECTRICAL SPECIFICATIONS

Logic Voltage: 5V DC (Arduino Nano)
High Voltage Rail: 30V - 40V DC (Adjustable via Boost Converter)
Max Voltage (Do not exceed): 50V DC
Current Draw (per piezo): ~10mA (pulse)
Current Draw (total array): ~80mA (all firing simultaneously)

FLUID DYNAMICS

Nozzle Size: ~0.3mm (27-gauge needle)
Ink Delivery: Gravity-fed manifold (sealed tank)
Leak Prevention: Surface tension at pinhole
Drop Formation: Piezo flex overcomes surface tension
Chamber Volume: ~2-5mL per nozzle

Prices Per Piece (PPP)

and where to get the parts

[IRL540N](#) 0.542\$ PP
[Piezo](#) 0.575\$ PP
[Booster](#) 6.17\$ PP and choose 250 watts
[Diodes](#) 0.0169\$ PP
[1K resistors \(already have\)](#) 0.027\$ PP
[74HC595](#) 0.169\$ PP
[Arduino Nano ATMEGA328P-1 \(choose that\)](#) 8.49\$ PP (Optional)
[27-gauge needle](#) 0.0338\$ PP (choose 50 pack)
[20MM EPTFE membrane](#) 0.1475\$ PP
Flexible membrane (use balloons) ~0.03\$ PP
[Various Tubing](#) 2.56\$ PM (Per Metre)

Quantity Per 8X Head (PX8-20H-01)

IRL540N X8
Piezo X8
Booster X1
Diodes X8
1K Resistors X8
74HC595 X1
Arduino Nano (Optional) X1
27-gauge needle X8
20MM EPTFE membrane X1
Flexible membrane X8

BOM for 8X Head (PX8-20H-01)

IRL540N: 4.336\$
Piezo: 4.6\$
Booster: 6.17\$
Diodes: 0.1352\$
1K Resistors: 0.216\$
74HC595: 0.169\$
Arduino nano (Optional): 8.49\$
27-gauge needles: 0.2704\$
20MM EPTFE membrane: 0.1475\$
Flexible membrane: ~0.24\$
Total with arduino: 24.7741\$
Total without arduino: 16.2841\$

Quantity Per 1X Head (PX1-20H-01)

IRL540N X1
Piezo X1
Booster X1
Diodes X1
1K Resistors X1
74HC595 X0 (don't need it if you're using just one nozzle)
Arduino Nano X1 (Optional)
27-gauge needle X1
20MM EPTFE membrane X1
Flexible membrane X1

BOM for 1X Head (PX1-20H-01)

IRL540N: 0.542\$

Piezo: 0.575\$

Booster: 6.17\$

Diodes: 0.0169\$

1K Resistors: 0.027\$

Arduino Nano: 8.49\$ (Optional)

27-gauge needle: 0.0338\$

20MM EPTFE membrane: 0.1475\$

Flexible membrane: ~0.03\$

Total with arduino: 16.0322\$

Total without arduino: 7.5422\$

Rules

- Never touch the output terminals with your fingers when it's on.
- Turn BOTH knobs on the booster fully counter-clockwise (to zero) before connecting the voltmeter
- Always unplug the 12V adapter and wait 10 seconds for the capacitor to drain before adjusting your wires.
- All grounds touch each other. Arduino GND, Battery Black, and MOSFET Source must be connected.
- The High Voltage Positive (+) is the only wire that should ever touch the Red wires of the piezos.
- The Arduino never sees high voltage. It only talks to the Gate via the 1k resistor.
- The diode only matters at high voltage. The silver stripe faces the red wire.
- Set the boost converter before connecting it. Measure it with a multimeter.
- Keep the ceramic dry. Only the brass side touches water.
- The 74HC595 is static sensitive so use anti static bags and dont touch if staticy
- If you don't have an anti-static bag for the 74HC595s, just wrap them in aluminum foil until you are ready to plug them in. The foil shorts all the pins together so no static can build up.
- If you print anything with pla, cover it in epoxy or clear nail polish to make it watertight

Check these before turning the booster on:

- The Nano stays on USB power (laptop or phone brick).
- High voltage and low voltage power are kept separate.
- The only connection they share is GND.
- Nano GND connects to the Boost Converter Negative.

Design rules:

- Print your nozzle chambers with a little recess (a shallow round hole) that matches the size of the plastic Luer Lock hub. The hub will sit perfectly flush, and a drop of glue will hold it forever.

The nano is optional because I am planning to put this nozzle inside a printer setup which will have an arduino anyway. If you want just the standalone nozzle, buy the arduino to control it.

For the nano that is in the parts list:

Go to Tools → Processor

If it says "ATmega328P", try that first.

If it fails to upload, change it to "ATmega328P (Old Bootloader)".

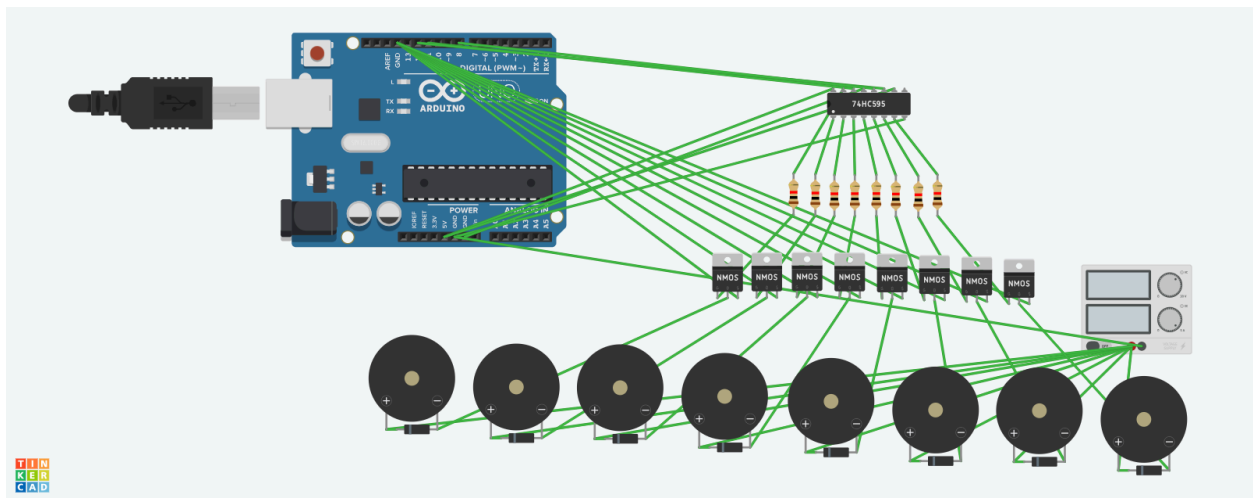
Try uploading again. It will work!

Wiring (W)

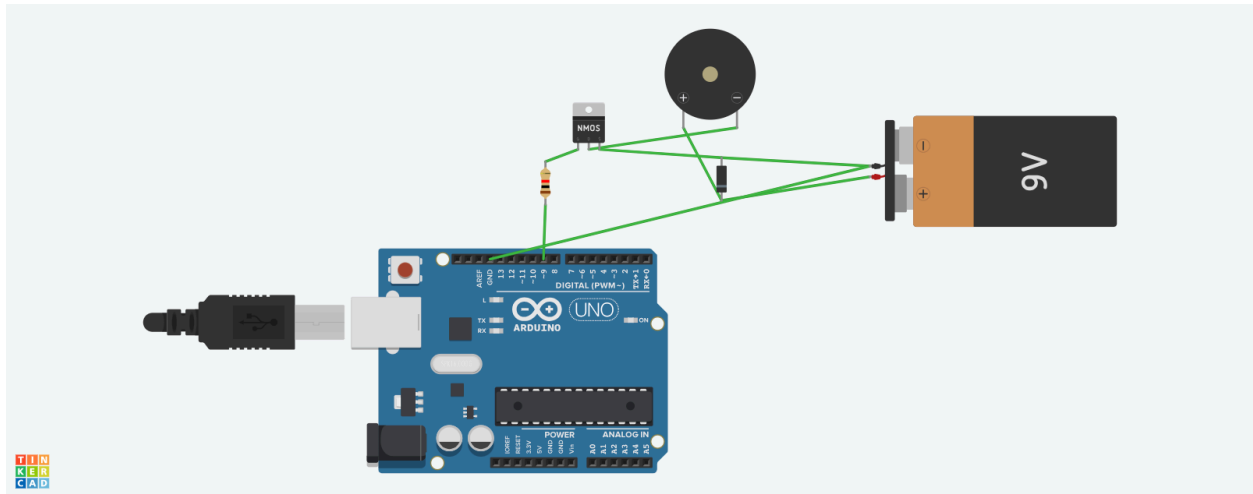
 Wiring diagram (for one piezo)

TinkerCad Designs:

8 Piezo Setup (works)

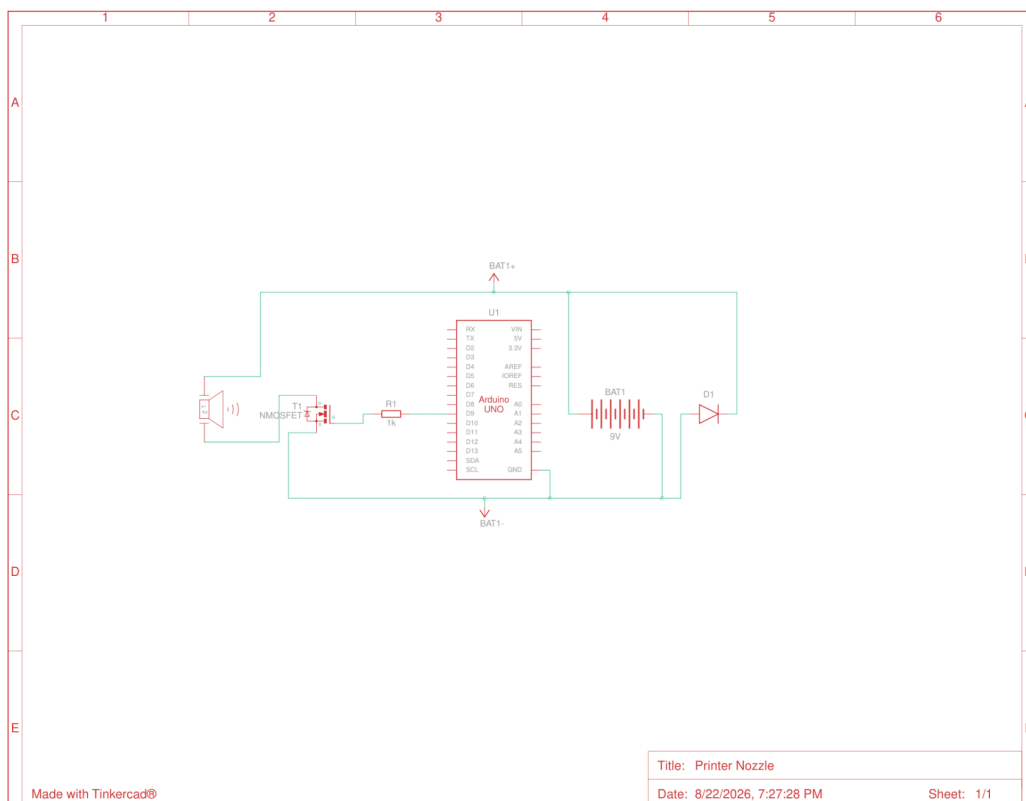


1 Piezo Setup (works too)

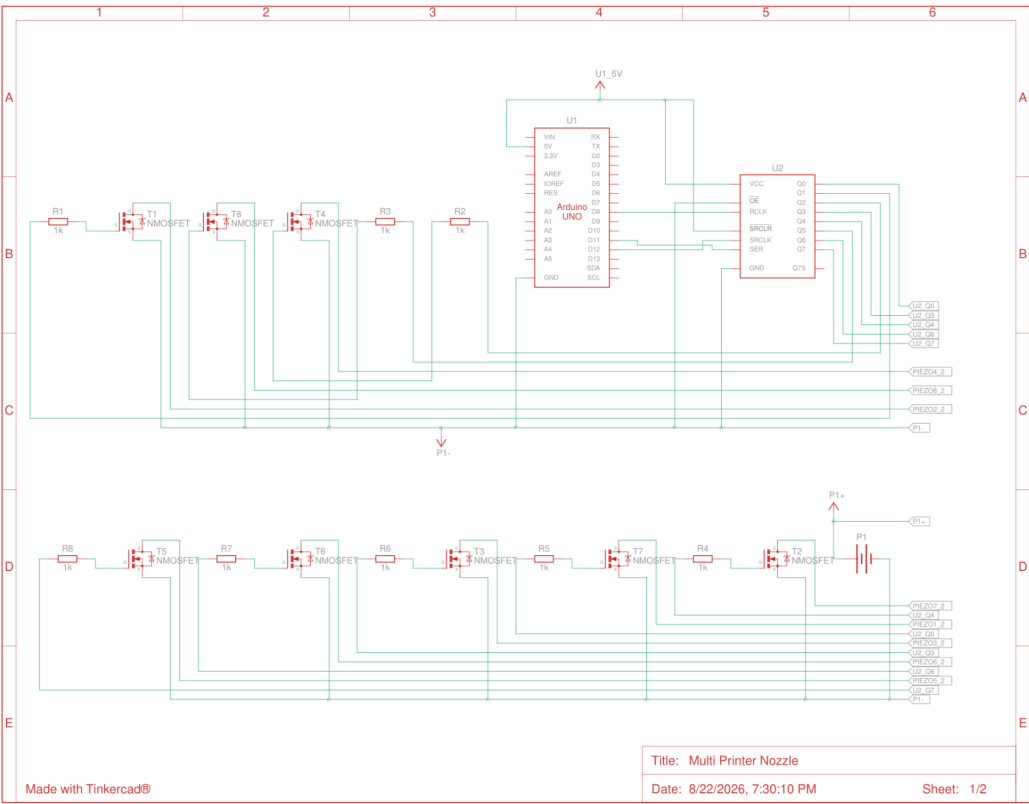


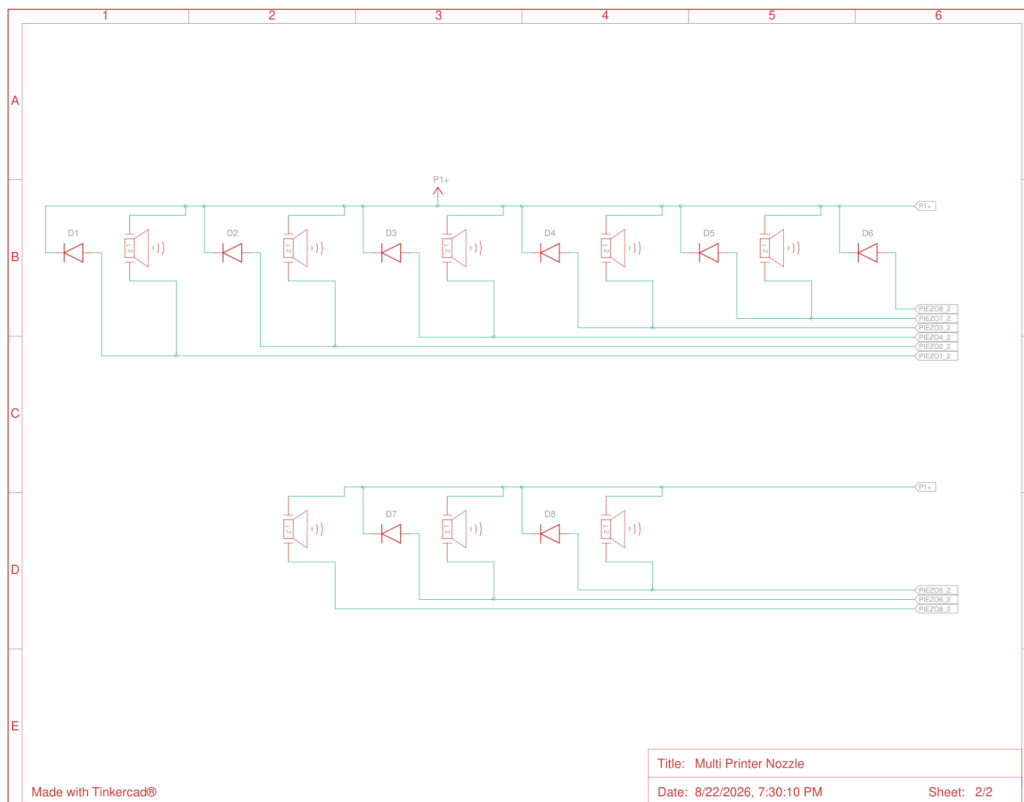
Schematics from TinkerCad:

1 Piezo Setup:

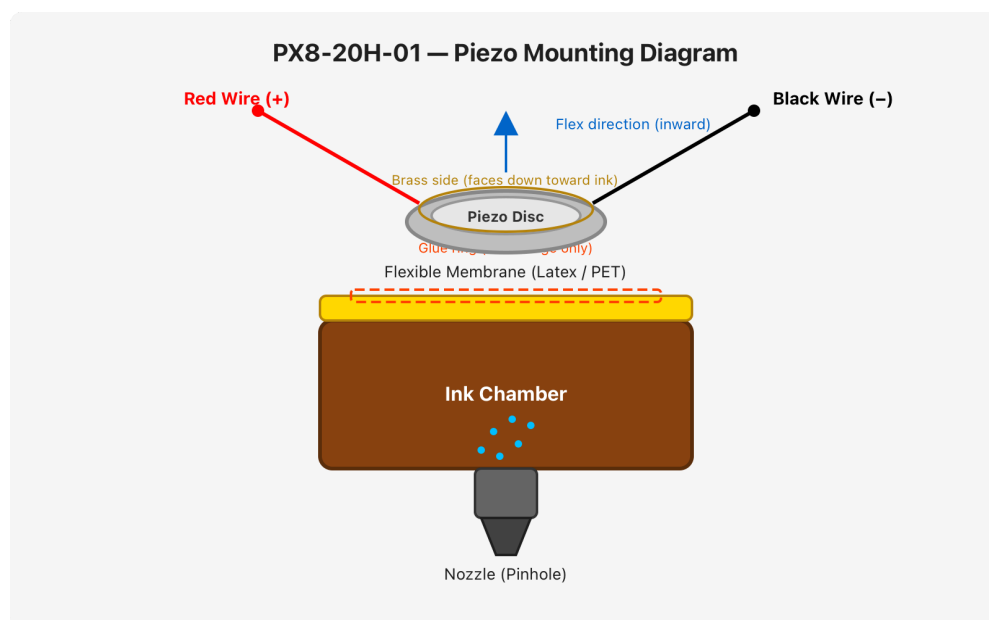


8 Piezo Setup:



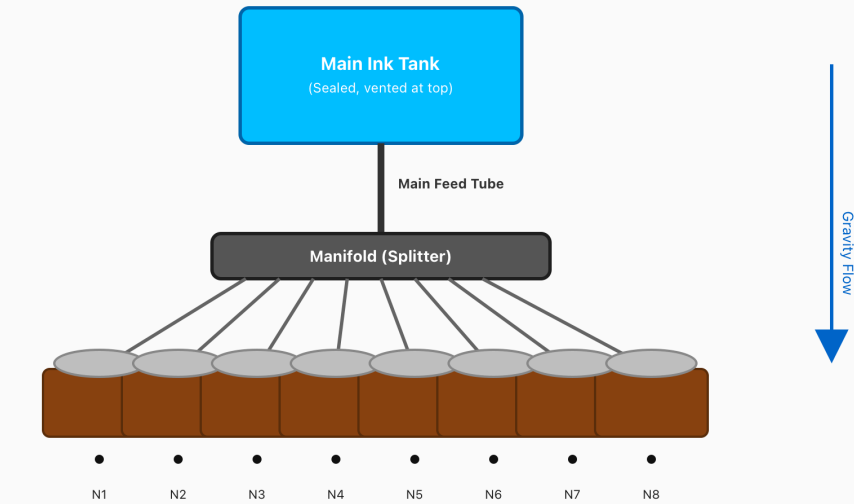


Physical installation



PX8-20H-01 — Ink Delivery System (8 Nozzle Array)

Gravity Fed Manifold with Capillary Inlets



Why doesn't it leak?

Surface tension at each pinhole acts like an invisible plug. The ink only escapes when the piezo flexes and forces it out.

PX8-20H-01 Passive Buoyancy Bubble Trap

